

CHEATSHEET · QUICK REFERENCE

Lab 5 — Out-of-Band Management, BIOS/UEFI, and Firmware

Inventory BIOS/UEFI, drive the ipmitool BMC command set, run the fwupd firmware workflow and walk the hot-swap pre-swap checklist.

BIOS / UEFI inventory

```
dmidecode -t bios
Full BIOS table

dmidecode -s bios-version
BIOS version string

[ -d /sys/firmware/efi ] && echo UEFI || echo legacy BIOS
Boot mode

efibootmgr -v
UEFI boot entries

efibootmgr -o 0002,0001,0003
Change boot order
```

IPMI / BMC command pattern

```
ipmitool -H bmc -U admin -P *** chassis status
Remote power / chassis state

ipmitool -H bmc -U admin -P *** power on
Remote power on/off/cycle

ipmitool -H bmc -U admin -P *** sol activate
Serial Over LAN console

ipmitool -H bmc -U admin -P *** sensor list
Temps, fan RPM, voltages

ipmitool -H bmc -U admin -P *** sel list
System Event Log (audit trail)

ipmitool -H bmc -U admin -P *** mc reset cold
Cold-reset the BMC
```

Firmware workflow with fwupd

```
fwupdmgr get-devices
Installed firmware inventory

fwupdmgr refresh
Pull latest LVFS metadata

fwupdmgr get-updates
Available upgrades

fwupdmgr update
Apply (back up + schedule + rollback first)
```

Hot-swap pre-swap checklist

Step	Check
1	Fault LED identifies the correct slot
2	Controller marks the disk Failed/Offline
3	For PSU/fan: confirm the redundant unit is healthy

Step	Check
4	Document serial of removed and replacement part
5	Watch dmesg -w / /proc/mdstat for clean re-add